

Institute of Actuaries of India

Subject CT8 – Financial Economics

December 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Utility functions which exhibit constant relative risk aversion are said to be iso-elastic. Iso elastic means that the elasticity of the marginal utility of wealth is constant with respect to wealth.

Example Power utility function or log utility function

$$U(w) = \frac{w^\gamma - 1}{\gamma}$$

$$U'(w) = w^{\gamma-1}$$

$$U''(w) = (\gamma-1) w^{\gamma-2}$$

$$A(w) = -\frac{U''(w)}{U'(w)} = -\frac{(\gamma-1)}{w}$$

$$A'(w) = \frac{(\gamma-1)}{w^2} < 0$$

$$\text{And } R(w) = w \left(-\frac{U''(w)}{U'(w)} \right) = -(\gamma-1)$$

$$R'(w) = 0$$

$$A'(w) < 0 \text{ and } R'(w) = 0$$

It exhibits declining absolute risk aversion and constant relative risk aversion. Hence it is iso elastic [3]

ii)

- a) Your friend is more risk averse than yours we have a lower risk aversion coefficient . for you

$$U(w) = \frac{w^\gamma - 1}{\gamma}$$

$$U'(w) = w^{\gamma-1}$$

$$U''(w) = (\gamma-1) w^{\gamma-2}$$

Substituting $\gamma = 1$

$$R(w) = A(w) = 0 \text{ i.e.}$$

For your friend $\gamma = 0.5$

$$A(w) = \frac{1}{2w} > 0$$

$$R(w) = \frac{1}{2} > 0$$

Hence your friend is strictly risk averse for all $w > 0$ [2]

- b) If your friend buys X then the expected utility

$$0.5(2((110)^{0.5} - 1)) + 2((92)^{0.5} - 1) = 18.08$$

However if he doesn't buy then expected utility is

$$2((100)^{0.5} - 1) = 18$$

Thus buying X has higher utility and hence he should buy X [3]

- iii) Utility theory is based on risk averse / risk seeker decreasing marginal utility theory based on total wealth. Prospect theory relates to how people make decisions when faced with risk and uncertainty. It is based on the concept of value defined in terms of gains and losses relative to a reference point [2]

iv)

- a) If m is the systematic risk portion of return, then:

$$m = \beta_{GDP} \Delta \% GDP + \beta_r \Delta \text{Interest rates}$$

$$m = 2.04(4.8\% - 3.5\%) - 1.90(7.80\% - 7.10\%)$$

$$m = 1.32\% \quad [2]$$

- b) The unsystematic is the return that occurs because of a firm specific factor such as the increase in market share.

If ϵ is the unsystematic risk portion of the return, then:

$$\epsilon = 0.36(27\% - 23\%) \quad \epsilon = 1.44\% \quad [2]$$

- c) The total return is the expected return, plus the two components of unexpected return: the systematic risk portion of return and the unsystematic portion.

So, the total return of the stock is:

$$R = R + m + \epsilon$$

$$R = 10.50\% + 1.32\% + 1.44\%$$

$$R = 13.26\% \quad [2]$$

[16 Marks]

Solution 2:

- i) VaR is the maximum potential loss on a portfolio over a given future time period with a degree of confidence where the latter is normally expressed as $1-p$

For continuous distribution

$$\text{VaR}(X) = -t \text{ where } P(X < t) = p$$

The problem with VaR is that it is often calculated assuming that investment returns are normally distributed. Portfolios exposed to credit risk or systematic bias may exhibit non – normal distribution. E.g Hedge funds

Also the other weakness of VaR is that it does not quantify the size of the tail [3]

- ii) We have to find $P(X < t) = 0.05$

$$X \sim N(5, 6^2)$$

Standardizing gives

$$P(Z < (t-5)/6) = \Phi((t-5)/6) = 0.05$$

From the tables $\Phi(-1.65) = 0.05$

$$\text{So } t = -4.9$$

Since t is a percentage investment return per annum, the 95% value at risk over one year on a 100 cr portfolio is $100 \times 4.9\% = 4.9$ cr.

This means that, we are 95% certain that we will not lose more than 4.9 cr over the next year [4]

iii) The mean return

$$E(x) = -7 \times 0.04 + 5.5 \times 0.96 = 5 \text{ i.e. } 5\% \text{ p.a.}$$

$$\text{Var}(X) = (5 - (-7))^2 \times 0.04 + (5 - 5.5)^2 \times 0.96 = 6\%$$

We first find t where $t = \max(x: P(X \leq x) \leq 0.05)$

$$P(X \leq -7) = 0 \text{ and } P(X \leq 5.5) = 0.04$$

Therefore $t = 5.5$

The expected shortfall in returns below 5.5% is given by

$$E(\max(5.5 - X, 0)) = \sum_{x < 5.5} (5.5 - x)P(X = x)$$

$$= 12.5 \times 0.04 = 0.5$$

On a portfolio of 10 cr the 95% Tail VaR is $10 \times 0.005 = 0.05$ cr. This means the expected reduction in profit below INR 5.5 cr is INR 500,000/- [4]

[11 Marks]

Solution 3:

i) We are given that

$\Delta t = 6$ months

$u/d = 4/3$

$q = 1/3$

Initial stock price (S_0) = 80

Strike price (K) = 85

Risk free continuously compounded rate = 5%

the risk-neutral probability of an up move is

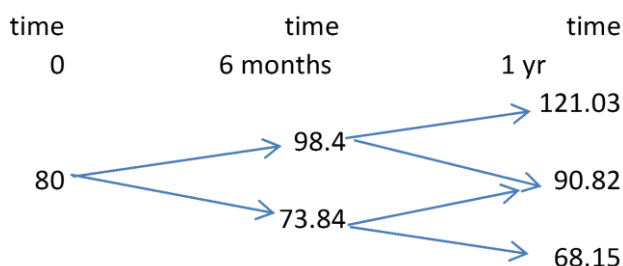
$$q = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{r\Delta t}/d - 1}{u/d - 1}$$

$$\frac{1}{3} = \frac{e^{0.05 \cdot 0.5}/d - 1}{4/3 - 1}$$

Calculating upto 3 decimal places we have

$$d = 0.923 \text{ and } u = 1.230$$

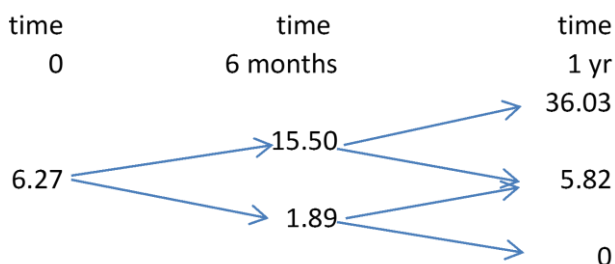
The two-period binomial tree for the stock price at $t = 0.5$ and $t = 1$ is given below.



Price of a one-year European call option with a strike of 85 (CI)

Maturity payoffs are $\max(S-85, 0)$

Working backwards through the tree with $V = \exp(-0.05 \cdot 0.5)[1/3 \cdot V_{up} + (1-1/3) \cdot V_{down}]$:



Therefore CI = 6.27

Price of a one-year American call option with a strike of 85 (CII)

Maturity payoffs are still $\max(S-85, 0)$

But in working backwards through the tree, we need to allow for early exercise option with V

$$= \max\{\exp(-0.05 \cdot 0.5)[1/3 \cdot V_{up} + (1-1/3) \cdot V_{down}], S-85\}.$$

We observe that at time 6 months

$V_u = 15.50$ (from tree calculated above)

$$\text{And } S_u - 85 = 98.4 - 85 = 13.4$$

Since $V_u > S_u - 85$, it is not optimal to exercise option early

Hence $V_u^{\text{American}} = \max\{\exp(-0.05 \cdot .5)[1/3 \cdot V_u + (1-1/3) \cdot V_{\text{down}}], S_u - 85\} = 15.50$

Which is similar to European option

similarly

$V_d^{\text{American}} = \max\{\exp(-0.05 \cdot .5)[1/3 \cdot V_u + (1-1/3) \cdot V_{\text{down}}], S_d - 85\} = 1.89$

similar to European option

therefore $CII = 6.27$

Thus, $CII - CI = 0$

Hence it is never optimal to exercise American Call option on a non-dividend paying stock.

[6]

ii) Expected price at node 1 from the tree (time 6 months)

$$E(S_1) = 1/3 \times 98.4 + (1-1/3) \times 73.84 = 82.02667$$

Check: the expectation should be

$$S_0 \exp(r \cdot \Delta t) = 80 \exp(0.05 \cdot .5) = 82.02521$$

These two expected values are very close. So the drift has been calibrated correctly.

Variance of S_1 from the tree is:

$$\begin{aligned} \text{Var}[S_1] &= E[S_1^2] - E[S_1]^2 \\ &= 1/3 \cdot 98.4^2 + (1-1/3) \cdot 73.84^2 - 82.02667^2 \\ &= 134.04247 \end{aligned}$$

Standard deviation of price at node 1 from the tree is

$$\begin{aligned} &\sqrt{134.04247} \\ &= 11.5777 \end{aligned}$$

Check: We expect to Standard deviation to be $80 \cdot \sqrt{[\exp(2\mu + \sigma^2) \Delta t] \cdot (\exp(\sigma^2 \Delta t) - 1)}$, where

volatility $\sigma = 20\%$ and

drift $\mu = r - \frac{1}{2} \sigma^2 = 3\%$,

$$\text{i.e. } 80 \cdot \sqrt{[\exp(2 \cdot .03 + .22) \cdot .5] \cdot (\exp(0.22 \cdot .5) - 1)} = \sqrt{135.9173} = 11.6583$$

These two values are also very close. Hence the tree has been calibrated appropriately

[4]

[10 Marks]

Solution 4:

- i) A Markov process is a particular type of stochastic process where only the present value of a variable is relevant for predicting the future. The past history of the variable and the way that the present has emerged from the past are irrelevant

[1]

- ii) The Markov property of stock prices is consistent with the weak form of market efficiency. This states that the present price of a stock impounds all the information contained in a record of past prices.

If the weak form of market efficiency were not true, technical analysts could make above average returns by interpreting charts of the past history of stock prices. There is very little evidence that they are in fact able to do this.

[3]

- iii) $E[M_{n+1} | S_0, \dots, S_n] = E[(S_n + X_{n+1})^2 - (n+1)\sigma^2 | S_0, \dots, S_n]$
 $= E[S_n^2 + 2S_nX_{n+1} + X_{n+1}^2 - (n+1)\sigma^2 | S_0, \dots, S_n]$
 $= S_n^2 + 2S_nE[X_{n+1} | S_0, \dots, S_n] + E[X_{n+1}^2 | S_0, \dots, S_n] - (n+1)\sigma^2$
 $= S_n^2 + \sigma^2 - (n+1)\sigma^2$
 $= M_n.$

[6]

[10 Marks]**Solution 5:**

- i) **Formula for delta**

Given that

$$V = S e^{-qT} \Phi(d_1) - K e^{-rT} \Phi(d_2)$$

$$\text{where } d_1, d_2 = \frac{\ln\left(\frac{S}{K}\right) + (r - q \pm 1/2\sigma^2)T}{\sigma\sqrt{T}} \quad \text{and}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Differentiating the formula given for V with respect to S , using the product rule for the second term, we get:

$$\Delta = \frac{\partial V}{\partial S} = e^{-qT} \Phi(d_1) + S e^{-qT} \phi(d_1) \frac{\partial d_1}{\partial S} - K e^{-rT} \phi(d_2) \frac{\partial d_2}{\partial S}$$

Here $\phi(\cdot)$ denotes the derivative of $\Phi(\cdot)$ ie it is the probability density function of the standard normal distribution, which is:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

From the definitions of d_1 and d_2 , we find that:

$$\frac{\partial d_1}{\partial S} = \frac{\partial d_2}{\partial S} = \frac{1}{S\sigma\sqrt{T}}$$

Putting these together and regrouping gives:

$$\Delta = \frac{\partial V}{\partial S} = e^{-qt}\Phi(d_1) + \frac{Se^{-qt}e^{-\frac{1}{2}d_1^2} - Ke^{-rt}e^{-\frac{1}{2}d_2^2}}{S\sigma\sqrt{T}}$$

For calculating the second term,

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln\left(\frac{S}{K}\right) + (r - q - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \quad \text{so that}$$

$$d_1 + d_2 = \frac{2 \ln\left(\frac{S}{K}\right) + 2(r - q)T}{\sigma\sqrt{T}} \quad \text{and}$$

$$d_1 - d_2 = \sigma\sqrt{T}$$

Multiplying these two equations gives us a difference of two squares:

$$d_1^2 - d_2^2 = 2 \ln\left(\frac{S}{K}\right) + 2(r - q)T$$

$$\frac{1}{2}d_1^2 - \frac{1}{2}d_2^2 = \ln\left(\frac{S}{K}\right) + (r - q)T$$

Exponentiating gives:

$$e^{\frac{1}{2}d_1^2} e^{-\frac{1}{2}d_2^2} = \frac{S}{K} e^{rt} e^{-qt}$$

Hence

$$Se^{-qt} e^{\frac{1}{2}d_1^2} = Ke^{-rt} e^{-\frac{1}{2}d_2^2}$$

$$\text{i.e } Se^{-qt} e^{\frac{1}{2}d_1^2} - Ke^{-rt} e^{\frac{1}{2}d_1^2} = 0$$

Therefore

$$\Delta = \frac{\partial V}{\partial S} = e^{-qT} \Phi(d_1)$$

[6]

ii) Formula for gamma

Finding the second derivative, starting from the formula in (i), is more straightforward:

$$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S} = e^{-qT} \phi(d_1) \frac{\partial d_1}{\partial S}$$

So

$$\Gamma = \left(\frac{e^{-qT} \phi(d_1)}{S \sigma \sqrt{T}} \right)$$

[2]

iii) Formula for speed

$$\frac{\partial \Gamma}{\partial S} = \frac{\partial}{\partial S} \left(\frac{e^{-qT} \phi(d_1)}{S \sigma \sqrt{T}} \right)$$

$$\text{Using } \phi(d_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2}$$

$$\frac{\partial \Gamma}{\partial S} = \frac{\partial}{\partial S} \left(\frac{e^{-qT} e^{-\frac{1}{2}d_1^2}}{S \sigma \sqrt{2\pi T}} \right)$$

$$= \left(\frac{e^{-qT} e^{-\frac{1}{2}d_1^2}}{\sigma \sqrt{2\pi T}} \right) \left(-\frac{1}{S^2} \right) + \frac{\partial}{\partial d_1} \left(\frac{e^{-qT} e^{-\frac{1}{2}d_1^2}}{S \sigma \sqrt{2\pi T}} \right) \frac{\partial d_1}{\partial S}$$

$$= \left(\frac{e^{-qT} e^{-\frac{1}{2}d_1^2}}{\sigma \sqrt{2\pi T}} \right) \left(-\frac{1}{S^2} \right) + \left(\frac{e^{-qT}}{S \sigma \sqrt{2\pi T}} \right) \left(e^{-\frac{1}{2}d_1^2} * (-d_1) \right) \left(\frac{1}{S \sigma \sqrt{T}} \right)$$

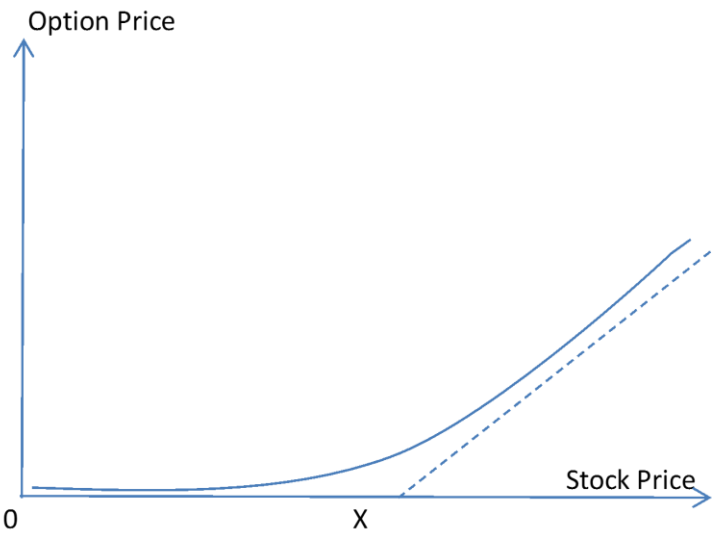
$$= -\left(\frac{1}{S} \right) \left(\frac{e^{-qT} e^{-\frac{1}{2}d_1^2}}{S \sigma \sqrt{2\pi T}} \right) \left[1 + \left(\frac{d_1}{\sigma \sqrt{T}} \right) \right]$$

$$= -\frac{\Gamma}{S} \left[1 + \left(\frac{d_1}{\sigma \sqrt{T}} \right) \right]$$

[3]

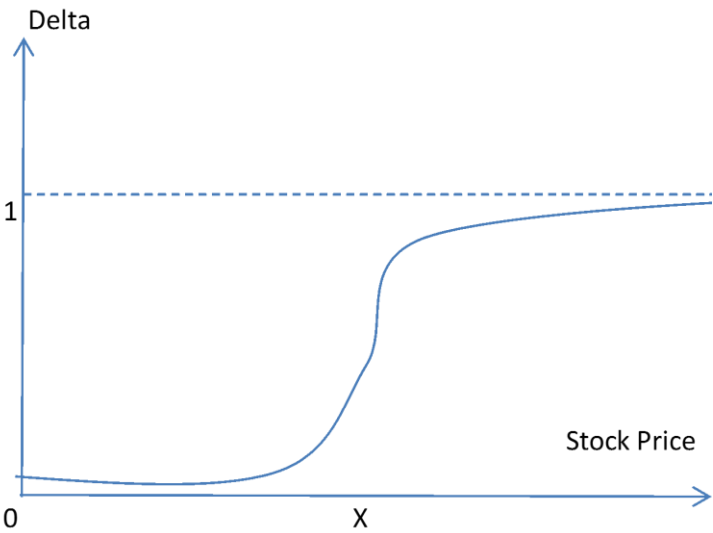
iv) graphs

a) price



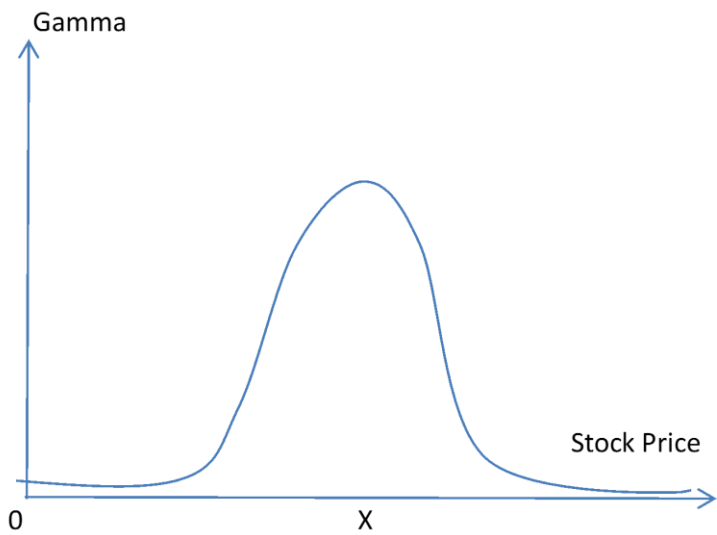
[1]

b) delta



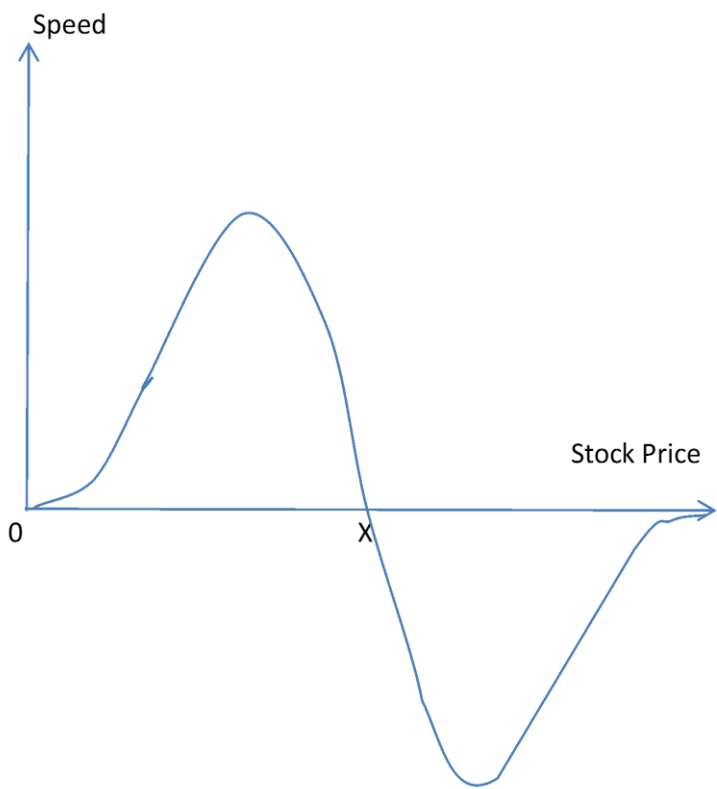
[1]

c) gamma



[1]

d) speed

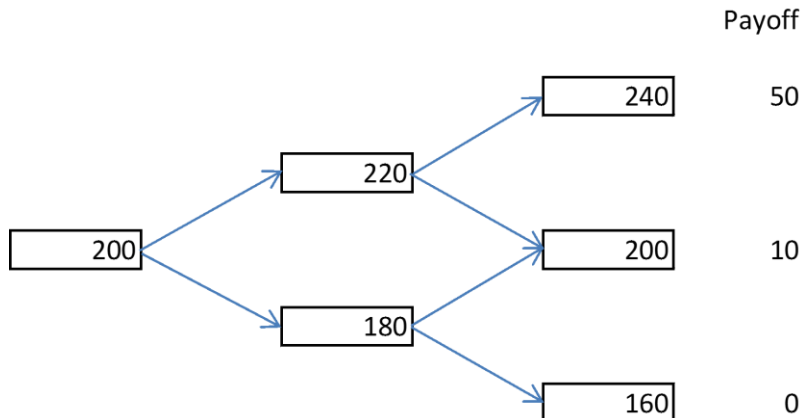


[1]

[15 Marks]

Solution 6:

i) Stock price movement is shown below:



First we calculate the risk-neutral probability of an upwards movement in the stock price from each state:

$$q(200) = \frac{1.02 * 200 - 180}{220 - 180} = 0.60$$

$$q(220) = \frac{1.02 * 220 - 200}{240 - 200} = 0.61$$

$$q(180) = \frac{1.02 * 180 - 160}{200 - 160} = 0.59$$

Then the option price V is:

$$V = \frac{[q(200) * q(220) * 50 + q(200) * (1 - q(220)) * 10 + (1 - q(200)) * q(180) * 10 + (1 - q(200)) * (1 - q(180)) * 0]}{1.02^2}$$

$$= 22.10688$$

[3]

ii)

a) Using the results from (i) we can calculate the values of the state-price deflator:

$$A(240) = \frac{q(200) * q(220)}{p^2 * (1 + r)^2} = \frac{.6 * .61}{.8^2 * (1.02)^2} = .54967$$

$$A(200) = \frac{q(200) * (1 - q(220)) + (1 - q(200)) * q(180)}{2 * p * (1 - p) * (1 + r)^2} = \frac{.6 * (1 - .61) + (1 - .6) * .59}{2 * .8 * .2 * (1.02)^2} = 1.41172$$

$$A(160) = \frac{(1 - q(200)) * (1 - q(180))}{(1 - p)^2 * (1 + r)^2} = \frac{(1 - .6) * (1 - .59)}{.2^2 * (1.02)^2} = 3.94079$$

Alternatively, we may proceed by calculating four deflators, one for each path:

$$\begin{aligned}
 A(uu) &= 0.54967, \\
 A(ud) &= 1.40571, \\
 A(du) &= 1.13412, \\
 A(dd) &= 3.94079
 \end{aligned}$$

[2]

- b) The option premium V can be calculated as:

$$V = E_P(A_2 V_2)$$

$$= [p^2 A(240) * 50 + 2 * p * (1-p) * A(200) * 10 + (1-p)^2 * A(160) * 0]$$

$$= [.8^2 A(240) * 50 + 2 * .8 * (1-.8) * A(200) * 10 + (1-.8)^2 * A(160) * 0]$$

$$= 22.10694$$

[2]

- c) This is the same answer as under part (i) as expected – under a given model the option price should not vary depending on how we evaluate the model [1]

iii)

- a) $A(240)$ would rise as the denominator decreases; $A(200)$ and $A(160)$ would shrink as the denominator rises [1]

- b) Overall the option price would remain unchanged as it does not depend on real-world probabilities [1]

[10 Marks]

Solution 7:

i)

- a) A process φ is *previsible* if it is known based on information up to but not including time t

[1]

- b) A portfolio (ϕ_t, ψ_t) is *self-financing* if ϕ_t, ψ_t are previsible and $dV_t = \phi_t dS_t + \psi_t dB_t$,

That is, at $t+dt$ there is no inflow or outflow of money necessary to make the value of the portfolio back up to $V(t+dt)$ ie the required change in the value of the portfolio over each instant of time is equal to the pure instantaneous investment gain

OR

A portfolio is self-financing if and only if changes in its value depend only on changes in the prices of the assets constituting the portfolio, i.e. there is no external cash amounts entering or leaving the portfolio.

Algebraically: If V_t denotes the value of the portfolio (ϕ_t, ψ_t) of two variables S_t and B_t respectively, then the portfolio is self financing if and only if

$$dV_t = \phi_t dS_t + \psi_t dB_t$$

[1]

- c) (Let us consider a derivative with random variable payoff X at time T .

A replicating strategy is a self-financing strategy (ϕ_t, ψ_t) , defined for $0 \leq t < U$, such that:

$$V(U) = \phi_U S_U + \psi_U B_U = X$$

In other words, for an initial investment of $V(0)$ at time 0, if we follow the self-financing portfolio strategy (ϕ_t, ψ_t) we will be able to reproduce the derivative payment without risk [1]

d) An investment market is *complete* if for every derivative in that market, there exists a replicating strategy for that derivative

[1]

ii) We are given

$$dS_t = \mu S_t dt + \sigma S_t dw_t$$

$$dB_t = rB_t dt$$

Applying Ito's Lemma in the real world:

$$d\left(\frac{B_t}{S_t}\right) = \frac{dB_t}{S_t} - \frac{B_t}{S_t^2} dS_t + \frac{1}{2} \frac{B_t}{S_t^3} dS_t^2$$

$$\text{Where } dS_t^2 = \sigma^2 S_t^2 dt$$

This, on substitution, reduces to

$$d\left(\frac{B_t}{S_t}\right) = \frac{rB_t dt}{S_t} - \frac{B_t}{S_t} (\mu dt + \sigma dw_t) + \sigma^2 \frac{B_t}{S_t} dt$$

$$= \frac{B_t}{S_t} (r - \mu + \sigma^2) dt - \frac{B_t}{S_t} \sigma dw_t$$

Hence $\frac{B_t}{S_t}$ is a martingale if it is driftless, i.e. if $\mu = r + \sigma^2$

[3]

iii) Let V_t be the value of the portfolio, i.e.

$$V_t = \phi_t S_t + \psi_t B_t$$

$$\text{Since } \phi_t = \frac{V_t - \psi_t B_t}{S_t} \Rightarrow V_t = S_t^2 + \psi_t B_t$$

Therefore

$$dV_t = 2S_t dS_t + (1/2) * 2 dS_t^2 + \psi_t dB_t + B_t d\psi_t \quad \text{_____ (A)}$$

$$\text{Where } dS_t^2 = \sigma^2 S_t^2 dt$$

However a self financing portfolio must be such that

$$dV_t = \phi_t dS_t + \psi_t dB_t \\ \Rightarrow dV_t = S_t dS_t + \psi_t dB_t \quad \text{_____ (B)}$$

From equation (A) and (B), we have

$$S_t dS_t + \psi_t dB_t = 2S_t dS_t + (1/2) * 2 dS_t^2 + \psi_t dB_t + B_t d\psi_t$$

$$0 = S_t dS_t + dS_t^2 + B_t d\psi_t$$

$$0 = S_t(\mu S_t dt + \sigma S_t dw_t) + \sigma^2 S_t^2 dt + B_t d\psi_t$$

$$= > B_t d\psi_t = - (S_t(\mu S_t dt + \sigma S_t dw_t) + \sigma^2 S_t^2 dt)$$

$$= - S_t^2(\mu + \sigma^2) dt - \sigma S_t^2 dw_t$$

Putting $\mu = r + \sigma^2$

$$B_t d\psi_t = - S_t^2[(r + 2\sigma^2) dt + \sigma dw_t]$$

$$d\psi_t = - (S_t^2/B_t)[(r + 2\sigma^2) dt + \sigma dw_t]$$

and ψ_t is the integral of this from 0 to t (since $\psi_0 = 0$).

[4]

[11 Marks]

Solution 8:

i) The equations defining the two models are similar:

$$\text{Vasicek: } dr(t) = \alpha(\mu - r(t))dt + \sigma d\hat{W}(t)$$

$$\text{Hull-White: } dr(t) = \alpha(\mu(t) - r(t))dt + \sigma d\hat{W}(t)$$

Both are:

- continuous-time Markov models of the short rate of interest
- Ito processes defined by a stochastic differential equation
- one-factor models (ie they incorporate only one source of randomness)
- usually defined in terms of a standard Brownian motion under the risk-neutral probability measure
- Both models imply that the short rate is mean-reverting
- Both models imply that the future short rate has a normal distribution
- Both models allow negative values for the short rate

The key difference is that the Vasicek model is time-homogeneous (with constant μ), whereas the Hull-White model is not (with time-dependent μ)

The Hull-White model has to be calibrated (ie a function has to be selected for $\mu(t)$) to match the current pattern of bond prices

Both models are mathematically tractable, although the Hull-White model is slightly more complicated algebraically

Hull-White can provide a better fit to historical data

[3]

ii) We know that

$$d(e^{at}r(t)) = e^{at}dr(t) + r(t)ae^{at}dt$$

$$= e^{at}((\theta(t) - ar(t))dt + \sigma d\hat{W}(t)) + r(t)ae^{at}dt$$

$$= e^{at}\theta(t)dt + \sigma e^{at}d\hat{W}(t)$$

[2]

iii) Solving SDE

$$\int_0^t d(e^{as}r(s)) = \int_0^t e^{as}\theta(s)ds + \sigma \int_0^t e^{as}d\hat{W}(s)$$

$$\therefore e^{at}r(t) = e^{a \cdot 0}r(0) + \int_0^t e^{as}\theta(s)ds + \sigma \int_0^t e^{as}d\hat{W}(s)$$

Dividing both sides by e^{at} we get

$$r(t) = e^{-at}r(0) + \int_0^t e^{-a(t-s)}\theta(s)ds + \sigma \int_0^t e^{-a(t-s)}d\hat{W}(s)$$

[3]

[8 Marks]

Solution 9:

i) A *credit event* is an event that will trigger the default of a bond and includes the following:

- failure to pay either capital or a coupon
- loss event (*i.e.* where the company says that it is not going to make a payment)
- bankruptcy
- rating downgrade of the bond by a rating agency such as Standard and Poor's or Moody's.

[2]

ii) Value of a risky bond is

$$B(t, T) = e^{-r(T-t)} [1 - (1-\delta)(1 - \exp(-\int_t^T \lambda_s ds))]$$

where B is the bond price, λ is the risk-neutral default rate, δ is the recovery rate, and r is the risk-free rate.

Using the formula,

$$88.5 = e^{-0.02(5)} * 100 * [1 - (1-0.65)(1 - \exp(-\int_0^5 \lambda ds))]$$

$$.885 = e^{-0.02(5)} (1 - .35(1 - e^{-.5\lambda}))$$

$$\Rightarrow \lambda = 72.24\%$$

[3]

iii)

a) The real cumulative default probability from time 0 to time 2

Under the Merton model,

$$p^{\text{Real}}(\text{default}) = P(FT < K),$$

where K is the default point, and F_t is the market value of the company's assets at time t and P^{Real} is the real cumulative default probability .

Hence

$$\begin{aligned}
 P^{\text{Real}}(\text{default}) &= \Phi\left(-\frac{\ln(F_{0/K}) + (\mu - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \quad (\text{recall Prob of default is } \Phi(-d_2)) \\
 &= \Phi\left(-\frac{\ln(100_{/75}) + (.07 - \frac{1}{2}(.3^2))2}{.3\sqrt{2}}\right) \\
 &= \Phi(-0.7959) \\
 &= 0.21304
 \end{aligned}$$

[2]

b) $Q^{\text{Risk Neutral}}(\text{default}) = Q(FT < K)$

(replacing μ by r and performing the same calculation we get)

$$\begin{aligned}
 Q^{\text{Risk Neutral}}(\text{default}) &= \Phi\left(-\frac{\ln(F_{0/K}) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
 &= \Phi\left(-\frac{\ln(100_{/75}) + (.02 - \frac{1}{2}(.3^2))2}{.3\sqrt{2}}\right) \\
 &= \Phi(-0.56022) \\
 &= 0.28766
 \end{aligned}$$

[2]

[9 Marks]
